Module 2

Time Value of Money & Interest Rate

September 3 – 8, 2025

Time value of money

Principal (P) and interest (r):

Let FV be the future value at time T (measured in years)

- simple interest: $FV = P(1 + rT)$
- compound interest: $FV = P(1 + r)^T$ (annually)
- compound frequency: $FV = P(1 + r/m)^{mT}$, m = 1, 2, 4, 12, 365
    - m = 1: annually
    - m = 2: semi-annually
    - m = 4: quarterly
    - m = 12: monthly
    - m = 365: daily (see day count convention below)
- continuous compounding:

  $FV = P(1 + r/m)^{mT} = P\, e^{mT \log(1+r/m)} \rightarrow P\, e^{rT}$ $(m \rightarrow \infty)$

Example:

P =	1000	r =	0.05		
Years\CompFreq	1	2	4	12	Cont
1	1050.00	1050.63	1050.95	1051.16	1051.27
3	1157.63	1159.69	1160.75	1161.47	1161.83
5	1276.28	1280.08	1282.04	1283.36	1284.03
7	1407.10	1412.97	1415.99	1418.04	1419.07
10	1628.89	1638.62	1643.62	1647.01	1648.72
20	2653.30	2685.06	2701.48	2712.64	2718.28
30	4321.94	4399.79	4440.21	4467.74	4481.69

- nominal interest rate vs effective interest rate (= equivalent annually compounding rate)

example:

for discrete compounding: $1 + r_e = (1 + r/m)^m$

4% quarterly compounding rate (nominal rate):

→ $(1 + 0.01)^4 - 1 = 4.06\%$ effective interest rate

4% continuous compounding rate (nominal rate):

→ $e^{0.04} - 1 = 4.08\%$ effective interest rate

Present Value (PV)

- discounting: $PV = CF\ (1 + r/m)^{-mT}$ ($= CF\ e^{-rT}$ if continuous compounding). CF = Cash Flow at time T
- discount factor: $DF = (1 + r/m)^{-mT}$ ($= e^{-rT}$ if continuous compounding)

  DF = PV of $1 at time T
- value of a bond: PV of all future cash flows

P = 1000	C = 100 (10%, annual)		maturity = 5Y
Discount Rate =	8.00%	annual compounding	
Time	Cash Flow	DF	PV
1	100	0.92593	92.59
2	100	0.85734	85.73
3	100	0.79383	79.38
4	100	0.73503	73.50
5	1100	0.68058	748.64
		Total PV	1079.85

- bond internal rate of return - yield to maturity (YTM): given market price of bond (PV), solving for r = YTM.

  For the bond above,

PV	YTM
1420.35	1.27%
1222.90	4.87%
1079.85	8.00%
1012.23	9.68%
915.01	12.38%

  → bond price is inversely related to its yield
- how does one count T? → day count convention:

30/360, 30/365, ACT/360, ACT/365, ACT/ACT (ACT = actual)

Determinants of bond price/yield:

- general interest rate level
- credit quality (likelihood of honoring debt obligations) of the bond issuer

Mortgage

Mortgage is loan on house purchase, repaid periodically

Fixed rate mortgage with monthly repayment:

- P = principal
- r = mortgage rate for fixed monthly payments
- M = monthly payment
- T = duration (e.g., 5, 15, 30 years)

$$P = M\sum_{i=1}^{n}\frac{1}{\left(1+\frac{r}{12}\right)^{i}}, \text{ n} = 12\text{T}$$

Example:

P = \$100,000, T = 30yr (n = 30 x 12 = 360)

r = 3% → M = \$421.6

r = 7% → M = \$665.3

Summary

- TVM ←→ interest rate r
- Interest rate r reflects
  - Market environment
  - Riskyness of the future cash flow

    → Value of future cash flows at a given time horizon are not equal!

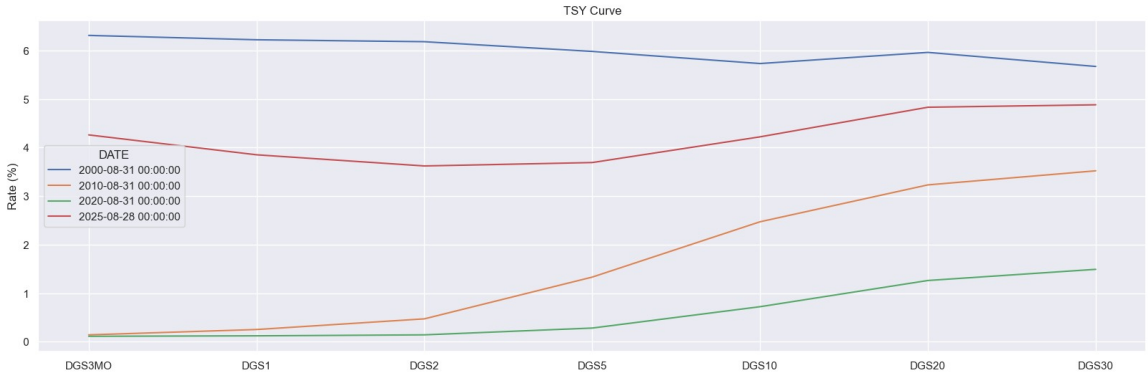
- Interest r is specified by compounding methods
  - Compounding frequency
  - Day count convention

Type of Rates

- treasury (TSY) rates
  - TSY bill (≤ 1Y), TSY notes (2Y - 10Y), TSY bonds (20, 30Y)
  - expecting government to not default on debts denominated in its own currency

    → credit risk free





(source: https://fred.stlouisfed.org/)

- fed fund rate

  → target interest rate set by the Federal Open Market Committee (FOMC)

- repo (repurchase agreement) rate

- LIBOR (London Interbank Offered Rate) – standard reference rate (till recent years)

  LIBOR replacements (more later in the course):

  USD:

  - SOFR - Secured Overnight Financing Rate
  - Term SOFR - series forward looking SOFR benchmark (administrated by CME, based on SOFR derivatives markets)
  - BSBY - Bloomberg Short Term Bank Yield Index
  - Ameribor - American Interbank Offered Rate: actual borrowing costs between mostly small and midsize banks

  EURO: ESTER (€STR, Euro Short Term Rate), EURIBOR

  UK Sterling: SONIA (Sterling Overnight Index Average)

  Swiss Franc: SARON (Swiss Average Rate Overnight)

  Japanese Yen: TORF (Tokyo Term Risk Free Rate), TONAR (Tokyo Overnight Average Rate)

- swap rates (more later in the course)
- OIS (overnight indexed swap) rates (more later in the course)

Terminlogy - basis point (bp): 1bp = 0.01%

Zero Rates

- Zero coupon bond:

  Let P(t, T) be the value of a T maturing zero coupon bond at time t

  P(T, T) = $N (notional).

  Let N = 1, t = 0

  - 1 – P(0, T) = interest paid on P
  - P = discount factor (DF) = PV of $1 at time T
  - $P = e^{-rT}$ → implied continuous compounded interest rate
  - Zero rates: rates implied from Ps

TSY bill quote on 2025-09-05 2:56 PM EST (source: Fidelity)

Cusip	Maturity Date	Price Bid	Price Ask	Yield Bid	Ask Yield	Quantity Bid(min)	Quantity Ask(min)
912797QU4	9/16/2025	99.905	99.906	4.338	4.292	30,000(100)	10,000(250)
912797RB5	10/7/2025	99.67	99.672	4.167	4.155	10,000(250)	15,000(1,000)
912797RW9	12/9/2025	99.005	99.01	3.984	3.968	10,000(100)	15,000(1,000)
912797RV1	3/5/2026	98.146	98.149	3.874	3.867	10,000(10)	27,699(100)
912797RF6	7/9/2026	97.012	97.025	3.671	3.656	5,000(200)	5,000(200)
912797RS8	9/3/2026	96.496	96.504	3.649	3.64	15,000(200)	80,000(300)

- Use zero rates to price a bond:

$$PV = \sum_{i=1}^{n} CF_i DF_i ,$$

For continuous compounding, $DF_i = e^{-r_i(t_i - t)}$

Example: consider a 5Y bond B with notional N = 1000 and annual coupon C = 6%

At t = 0:

N = 1000	C = 60 (6%, annual)		maturity = 5Y	
Discount Rate:		curve, continuous compounding		
Time	Cash Flow	Zero Rate	DF	PV
1	60	5.500%	0.94649	56.79
2	60	5.750%	0.89137	53.48
3	60	6.000%	0.83527	50.12
4	60	6.500%	0.77105	46.26
5	1060	7.000%	0.70469	746.97
			Total PV	953.62

What if the traded price of the above bond is different from 953.62?

Suppose the zero coupon bonds also have notional 1000. They are traded at the prices indicated below. Take the below position it will create an equivalent cash flow of 100 unit of the above coupon paying bond:

Maturity	Price	Position
1	946.49	6.00
2	891.37	6.00
3	835.27	6.00
4	771.05	6.00
5	704.69	106.00

→ "synthetic bond" → "replicate" B

→ risk free profit if the traded bond price B $\neq$ 953.62:

- B > 953.62: long synthetic, short B
- B < 953.62: short synthetic, long B

At maturity pocket the difference (assume "ideal market")

→ "arbitrage"

- $P = e^{-rT}$ → $\frac{\partial P}{\partial r} = -Te^{-rT} = -T \times P$

→ bond price sensitivity to the interest rate change increases with maturity

Determining Zero Rates (yield curve building)

- Bootstrapping:
  - use instruments with closest maturities to get short dated rates;
  - from the rates thus obtained and the price of the next maturity instrument to obtain the rate for the next period;
  - repeat the process
- Example: bootstrapping TSY curve (Hull: Chapter 4, p84 - 86)
- Interpolation, extrapolation method: piecewise flat, linear, spline

  Pitfall of spline method

Forward Rates

- Implied interest rate (r_{12}) to be paid/received at a future time interval [T_1 , T_2] from today's yield curve

  Assume continuous compounded zero rate for T_1 and T_2 are r_1 and r_2 , then

  $e^{r_1 T_1} e^{r_{12}(T_2 - T_1)} = e^{r_2 T_2}$ → $r_{12} = \frac{r_2 T_2 - r_1 T_1}{T_2 - T_1}$

  More general:

$$P(0,T_1)P(T_1,T_2) = P(0,T_2) \;\rightarrow\; DF(T_1,T_2) = P(T_1,T_2) = {P(0,T_2)}\big/{P(0,T_1)}$$

- Forward rate agreement (FRA): legally binding agreement on forward interest rate payment.

  Example:

  N = payment notional; $0 < T_1 , T_2$ = starting, ending period (usually $[T_1 , T_2] = 3$ or 6 month); f_{T_1,T_2} = FRA rate; reference floating rate = RFR:

  Payoff at T_1 = N (RFR – f_{T_1,T_2})($T_2 – T_1$),

  At t = 0, \$FRA = 0 → $f_{T_1,T_2} = r_{12}$

  If not, one may arbitrage (i.e., create positive \$payoff from a \$0 initial position) the mispricing in FRA rate.

  For simplicity assume interest on loans are paid at the end of the loan periods and the interest on the loan compounds with the same frequency as the FRA tenor ($T_2 – T_1$). \$1 lent at time 0, redeemed at T_1 and T_2 will thus grow to $P^{-1}(0,T_1)$ and $P^{-1}(0,T_2)$ respectively. If one lends a T_1 loan and then rolls the fund at T_1 to a T_2 loan following the FRA rate f_{T_1,T_2}, the terminal value at T_2 will be

  $$P^{-1}(0,T_1)\left(1+f_{T_1,T_2}(T_2-T_1)\right)$$

  - If $f_{T_1,T_2} > r_{12}$ then at t = 0 one can borrow a T_2 loan at r_2 to finance the above position. The terminal value of the entire position at T_2 will be

    $$P^{-1}(0,T_1)\left(1+f_{T_1,T_2}(T_2-T_1)\right)-P^{-1}(0,T_2) \;>\; P^{-1}(0,T_1)\left(1+r_{12}(T_2-T_1)\right)-P^{-1}(0,T_2)$$

    $$= P^{-1}(0,T_1)\,{P(0,T_1)}\big/{P(0,T_2)}-P^{-1}(0,T_2)$$

    $$= 0$$

  - If $f_{T_1,T_2} < r_{12}$: at t = 0 borrow T_1 loan and enter a $[T_1 , T_2]$ FRA paying f_{T_1,T_2}, lend out the fund in a T_2 loan. The terminal value of the entire position at T_2 will be

    $$P^{-1}(0,T_2)-P^{-1}(0,T_1)\left(1+f_{T_1,T_2}(T_2-T_1)\right) > 0$$

- Value of FRA at t > 0:

  See next lecture

- Value a bond with floating rate coupons (FRN = floating rate note)

Notional = N, floating coupon rate $r_{i,\,i+1}$ set at t_i , coupon payments (= $Nr_{i,i+1}t_{i,i+1}$) made at time t_{i+1}, i = 0, 1, …, n – 1,

$$FRN = N\sum_{i=0}^{n-1} r_{i,i+1}t_{i,i+1}DF(0,t_{i+1}) + N \times DF(0,t_n)$$

If discount factors and coupon payment rates are determined by the same yield curve, and compounding frequency and coupon payment frequency coincide,

$$DF(0,t_i) = P(0,t_i),\ DF(t_i,t_{i+1}) = \frac{1}{1+r_{i,i+1}t_{i,i+1}} = \frac{P(0,t_{i+1})}{P(0,t_i)}$$

$$FRN = N\sum_{i=0}^{n-1}\left(\frac{P(0,t_i)}{P(0,t_{i+1})} - 1\right)P(0,t_{i+1}) + N \times P(0,t_n)$$

$$= N\sum_{i=0}^{n-1}\left(P(0,t_i) - P(0,t_{i+1})\right) + N \times P(0,t_n)$$

$$= N\left(P(0,0) - P(0,t_n)\right) + N \times P(0,t_n)$$

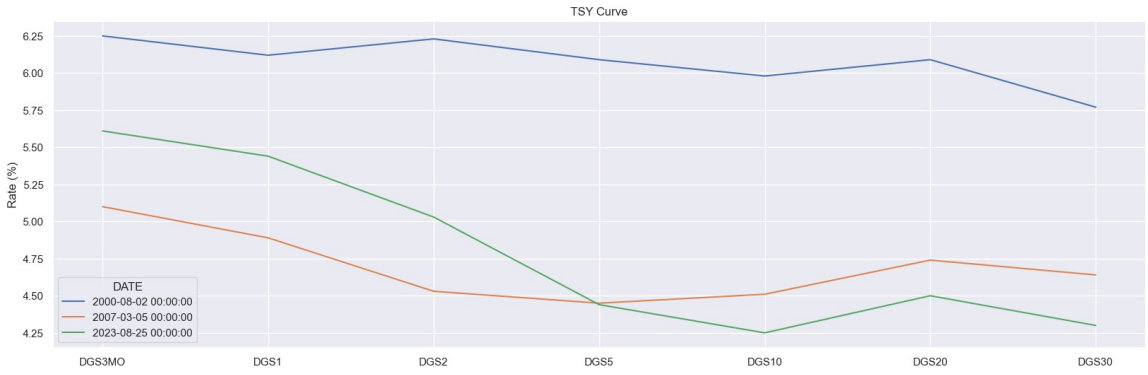
FRN = N (at rate reset date)

Theory of Term Structure of Interest Rate *

Term structure of interest rates: interest rate as function of time (maturity)

- Determinants of the shape of the term structure
  - expectations
  - risk preference
  - supply and demand
  - "convexity bias"
- Theories of term structure of interest rates
  - expectation theory
    - pure expectation
      - implied forward rate is unbiased estimate of market expectation of future interest rate

        - but what about risk – bond price risk, reinvest risk
    - liquidity preference theory
        - implied forward rate is NOT unbiased estimate of market expectation of future interest rate
        - expectation + liquidity premium
    - preferred habitat theory
        - market participants live in there preferred sections of yield curve
        - when supply-demand makes their sections unattractive they require risk premium to migrate
  - market segmentation theory
  - ….
- Implication of inverted yield curve
  - short term bonds → reinvestment risk
  - expecting worsening economy → prefer long term bond to sit out recession
    → demand higher yield for short term bond
  - predicted 2008 – 2009 recession
  - inversion in 2023 → ??



(source: https://fred.stlouisfed.org/)

Actual rates used by corporations and market participants

- yield curve used by banks for valuing derivatives:

  OIS, ("IBOR")/term SOFR, swaps

- rate vs discount factor:

  DF is independent from compounding and day count convention

- funding curve

- credit spread

A summary of recent developments:

"Interest Rate Instruments and Market Conventions Guide - Post LIBOR edition", Marc Henrard, 2025, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5099269